

■ Multi-Agent AI System Notes + Learning Roadmap

This document contains both the Multi-Agent AI architecture notes and a structured roadmap for learning AI step by step, from foundations to multi-agent systems.

■ Recap: Multi-Agent AI Architecture

AI technologies can be organized into three layers:

- **Core Agents:** Specialist AIs like Generative, Multimodal, VLA, Domain-specific.
- **System Intelligence:** Coordinators like Agentic AI, Multi-Agent Systems, Embodied AI.
- **Infrastructure & Governance:** Support layers like Edge AI, Long-context, Infrastructure, Explainability, Green AI, Personalization, Robotics integration.

■■ Step-by-Step AI Learning Roadmap

1. Foundations (Math & Programming)

- Math: Linear algebra, probability, statistics, calculus
- Python: NumPy, Pandas, Matplotlib
- Algorithms: Basics of optimization and search
- Goal: Comfort with the 'language' of AI

2. Core Machine Learning (ML)

- Supervised learning: regression, classification
- Unsupervised learning: clustering, PCA
- Model evaluation: accuracy, precision/recall, F1
- Libraries: scikit-learn, XGBoost
- Goal: Learn how machines detect patterns from data

3. Deep Learning (DL)

- Neural networks basics: perceptron, activation
- Architectures: CNNs, RNNs, Transformers
- Frameworks: TensorFlow, PyTorch
- Goal: Understand how deep nets power modern AI

4. AI Components (Skills for Agents)

- Computer Vision: image recognition, object detection
- NLP: classification, summarization, translation
- Reinforcement Learning: agents acting in environments
- Speech AI: speech-to-text, text-to-speech
- Goal: Collect building blocks that agents will use

5. Agents (Combining Components)

- Single-agent systems: chatbots, simple bots
- Autonomous agents: AutoGPT, copilots
- Goal: Build AI that plans and acts, not just classifies

6. Multi-Agent Systems (Combining Agents)

- Study coordination, communication, cooperation
- Frameworks: Microsoft AutoGen, LangChain multi-agent
- Goal: Teams of AIs working together like humans

7. Real-World Integration

- Edge AI for devices and robots
- Robotics with embodied AI

- Explainability, ethics, safety
- Goal: From lab to production systems

8. Specialization (Choose Your Path)

- Robotics & autonomous systems
- Generative AI & multimodal models
- Domain AI: healthcare, law, finance
- Goal: Deep expertise in chosen field

■ Key principle: Start with basics → build components → combine into agents → scale into multi-agent systems. This path ensures you can understand and design intelligent systems from the ground up.